Compiled by Maximus Pung Jun
O-Level Additional Mathematics (4049)
2026

2026 A Math Tangents Through a Point: Solutions with comments

1 Given that $y=x^3+2x^2-1$, find the equations of all the tangents to the curve that pass through the point $(-1.6,\ 0.6)$. [10]

[A Math/Tangents]

Solution		Comments
[10]	$\frac{dy}{dx}=3x^2+4x$ Point of contact $(a,\ a^3+2a^2-1)$: $m=3a^2+4a$ $y=(3a^2+4a)\,x+c$ $a^3+2a^2-1=(3a^2+4a)\,a+c$ $a^3+2a^2-1=3a^3+4a^2+c$ $c=-2a^3-2a^2-1$ $\therefore y=(3a^2+4a)\,x-2a^3-2a^2-1$ Through $(-1.6,\ 0.6)$: $0.6=(3a^2+4a)\,(-1.6)-2a^3-2a^2-1$ $0.6=-4.8a^2-6.4a-2a^3-2a^2-1$ $0.6=-2a^3-6.8a^2-6.4a-1$ $2a^3+6.8a^2+6.4a+1.6=0$ $5a^3+17a^2+16a+4=0 \quad (\times\frac{5}{2})$ $f(a)=5a^3+17a^2+16a+4$: $f(-2)=5(-8)+17(4)+16(-2)+4$ $=-40+68-32+4$ $=0$ $\therefore (a+2)$ is a factor	

$5a^3+17a^2+16a+4=(a+2)\,(5a^2+ka+2)$

coeff. of a^2: $17=k+10$

$k=7$

$5a^3+17a^2+16a+4=(a+2)\,(5a^2+7a+2)$

$=(a+2)(a+1)(5a+2)$

$(a+2)(a+1)(5a+2)=0$

$\therefore a=-2,\ a=-1,\ a=-\frac{2}{5}$

$a=-2$:

$y=(-2)^3+2(-2)^2-1$

$=-1$

$m=3(-2)^2+4(-2)$

$=4$

$y-(-1)=4(x+2)$

$\therefore y=4x+7$

$a=-1$:

$y=(-1)^3+2(-1)^2-1$

$=0$

$m=3(-1)^2+4(-1)$

$=-1$

$y-0=-1(x+1)$

$\therefore y=-x-1$

$a=-\frac{2}{5}$:

$y=\left(-\frac{2}{5}\right)^3+2\left(-\frac{2}{5}\right)^2-1$

$=-\frac{8}{125}+\frac{40}{125}-\frac{125}{125}$

$=-\frac{93}{125}$

$m=3\left(\frac{4}{25}\right)+4\left(-\frac{2}{5}\right)$

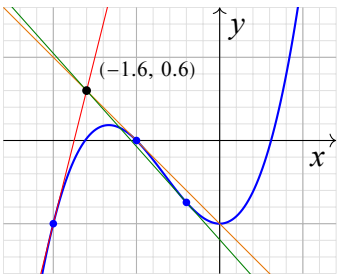
$=\frac{12}{25}-\frac{40}{25}$

$=-\frac{28}{25}$

$y+\frac{93}{125}=-\frac{28}{25}\left(x+\frac{2}{5}\right)$

$y=-\frac{28}{25}x-\frac{56}{125}-\frac{93}{125}$

$\therefore y=-\frac{28}{25}x-\frac{149}{125}$



Alternative method:

Point of contact $(a,\ a^3+2a^2-1)$:

$$\text{Gradient} = \frac{(a^3+2a^2-1)-0.6}{a-(-1.6)}$$

$$= \frac{a^3+2a^2-1.6}{a+1.6}$$

$$\frac{a^3+2a^2-1.6}{a+1.6} = 3a^2+4a$$

$$a^3+2a^2-1.6 = (3a^2+4a)\ (a+1.6)$$

$$a^3+2a^2-1.6 = 3a^3+4.8a^2+4a^2+6.4a$$

$$a^3+2a^2-1.6 = 3a^3+8.8a^2+6.4a$$

$$2a^3+6.8a^2+6.4a+1.6=0$$

$$5a^3+17a^2+16a+4=0 \quad \left(\times \tfrac{5}{2}\right)$$

$\therefore y=4x+7, \quad y=-x-1, \quad y=-\frac{28}{25}x-\frac{149}{125}$

END OF SOLUTIONS